

PMCS Assignment 1 – Detailed Solutions

Overview

This report provides detailed, step-by-step solutions to all four questions in the "PMCS Assignment-1" document on queueing systems and Markov processes. The derivations rely on standard results from M/M/1, M/G/1 queues, busy period analysis, and Jackson networks as presented in the attached textbooks by Mitrani, Ross, and Harchol-Balter.[1][2][3]

1. M/M/1/1 Queue with Two Classes

1.1 Model description

- Single server, system capacity 1 (loss system – Erlang loss type).
- Two independent Poisson arrival streams:
 - Class 1: rate λ_1 .
 - Class 2: rate λ_2 . [1]
- Service times are exponential and class-dependent:
 - Class 1: rate μ_1 (mean $1/\mu_1$).
 - Class 2: rate μ_2 . [1]
- Arrivals finding the server busy are lost (no queue).

Define the traffic parameters

$$\rho_i = \frac{\lambda_i}{\mu_i}, \quad i = 1, 2.$$

This matches the standard notion of offered load in single-server systems. [3]

1.2 CTMC and stationary distribution

The natural state descriptor is just the occupation of the server:

- State 0: server idle.
- State 1: server busy with a class-1 job.
- State 2: server busy with a class-2 job.[1]

Transition rates:

- From 0:
 - $0 \rightarrow 1$ at rate λ_1 .
 - $0 \rightarrow 2$ at rate λ_2 .
- From 1:
 - $1 \rightarrow 0$ at rate μ_1 (service completion).
 - Arrivals of either class are blocked and cause no state change.
- From 2:
 - $2 \rightarrow 0$ at rate μ_2 .

Let π_0, π_1, π_2 denote the stationary probabilities. Global balance gives

$$\lambda_1 \pi_0 = \mu_1 \pi_1, \quad \lambda_2 \pi_0 = \mu_2 \pi_2,$$

plus normalization

$$\pi_0 + \pi_1 + \pi_2 = 1.$$

Solving,

$$\pi_1 = \frac{\lambda_1}{\mu_1} \pi_0 = \rho_1 \pi_0, \quad \pi_2 = \frac{\lambda_2}{\mu_2} \pi_0 = \rho_2 \pi_0.$$

Hence

$$\pi_0(1 + \rho_1 + \rho_2) = 1 \Rightarrow \pi_0 = \frac{1}{1 + \rho_1 + \rho_2}, \quad \pi_i = \frac{\rho_i}{1 + \rho_1 + \rho_2},$$

This CTMC stationary computation is standard for finite-state birth-death like processes.[2]

1.3 Long-run fraction of time server is occupied by each class

By definition, the long-run fraction of time the server is occupied by class- i jobs is exactly π_i , since these are the stationary probabilities of the states in which the server is serving class i . [3]

Thus

$\text{Fraction of time server is serving class } i = \pi_i = \frac{\lambda_i/\mu_i}{1 + \lambda_1/\mu_1 + \lambda_2/\mu_2}$
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Sometimes it is useful to condition on the server being busy; then

$$P(\text{class } i \mid \text{server busy}) = \frac{\pi_i}{\pi_1 + \pi_2} = \frac{\rho_i}{\rho_1 + \rho_2}, \quad i = 1, 2.$$

1.4 Time-average vs ensemble-average occupation probabilities

Let $X(t)$ be the CTMC state at time t . For an ergodic CTMC (finite state, irreducible, positive recurrent), the time-average fraction of time spent in state j along almost every sample path equals the stationary probability π_j . [2]

Time-average occupation probability for class i :

$$\hat{p}_i = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \mathbf{1}\{X(t) = i\} dt.$$

Ensemble-average occupation probability (probability at an arbitrary time):

$$\bar{p}_i = \lim_{t \rightarrow \infty} P\{X(t) = i\} = \pi_i.$$

By the ergodic theorem for CTMCs, $\hat{p}_i = \bar{p}_i = \pi_i$ with probability 1. [2]

So time-average and ensemble-average occupation probabilities for each class are equal, and both are given by the π_i above.

2. Single-Server Queue with Setup Time

2.1 Model description

- Single server, infinite buffer (M/G/1 queue with modifications).
- Poisson arrivals with rate λ . [1]
- Service times i.i.d. with general distribution G , mean $E[S]$, finite second moment (needed for standard M/G/1 results). [3]
- Server policy:
 - When the system becomes empty, server immediately turns off.
 - Upon the next arrival to an empty system, the server incurs a setup time U (independent of arrivals and S) before starting service.
- Assume $E[U] < \infty$ and traffic intensity $\rho = \lambda E[S] < 1$ so that the system is stable. [3][1]

Goal: find long-run fractions of time the server is

- (a) busy serving jobs,
- (b) in setup,
- (c) idle (powered off).

2.2 Cycle decomposition

A standard technique for such systems is to decompose time into **cycles** between successive epochs at which the system has just become empty and the server has turned off. [2][3]

Define a cycle as:

1. Start: just after a job departure that leaves the system empty and the server turned off.
2. Then:
 - **Idle-off period:** wait for the next arrival. Because arrivals are Poisson with rate λ , this duration I is exponential with mean $E[I] = 1/\lambda$. [3]
 - **Setup period:** upon that arrival, a setup time U is incurred; mean $E[U]$.
 - **Busy period:** after setup finishes, the server works continuously until the system becomes empty again; denote

this busy period by B .

3. End of cycle: at the moment the system becomes empty again (server just turns off after the final service completion).

The idle-off, setup, and busy parts are mutually independent because arrivals, service times, and setup times are independent processes.[3]

Thus the cycle length is

$$C = I + U + B,$$

with

$$E[C] = E[I] + E[U] + E[B].$$

2.3 Mean busy period in an M/G/1 queue

If we ignore setup and just consider a standard M/G/1 queue with arrival rate λ and generic service S , a **busy period** is defined as the time from when the system changes from 0 to 1 job (an arrival to an empty system), until the time it next becomes empty.[3]

For an M/G/1 queue, it is classical that

$$E[B] = \frac{E[S]}{1 - \rho}, \quad \rho = \lambda E[S].$$

This can be derived either via a branching argument or via the simple fixed-point equation

$$E[B] = E[S] + \lambda E[B] E[S],$$

which rearranges to the expression above.[2][3]

Therefore in this problem

$$E[B] = \frac{E[S]}{1 - \lambda E[S]}.$$

The other components have means

$$E[I] = \frac{1}{\lambda}, \quad E[U] = E[U].$$

Hence the mean cycle length is

$$E[C] = \frac{1}{\lambda} + E[U] + \frac{E[S]}{1 - \lambda E[S]}.$$

2.4 Long-run fractions of time in each server state

By the **renewal-reward theorem**, the long-run fraction of time the system spends in a set of states equals the expected time spent in those states during a typical cycle divided by the expected cycle length.[2][3]

- Time serving jobs per cycle: $E[B]$.
- Time in setup per cycle: $E[U]$.
- Time idle (powered off) per cycle: $E[I] = 1/\lambda$.

Thus the required fractions are:

1. Busy serving jobs

$$\text{Fraction busy} = \frac{E[B]}{E[C]} = \frac{\frac{E[S]}{1 - \lambda E[S]}}{\frac{1}{\lambda} + E[U] + \frac{E[S]}{1 - \lambda E[S]}}.$$

2. In setup

$$\text{Fraction in setup} = \frac{E[U]}{E[C]} = \frac{E[U]}{\frac{1}{\lambda} + E[U] + \frac{E[S]}{1 - \lambda E[S]}}.$$

3. Idle (powered off)

$$\text{Fraction idle} = \frac{E[I]}{E[C]} = \frac{\frac{1}{\lambda}}{\frac{1}{\lambda} + E[U] + \frac{E[S]}{1 - \lambda E[S]}}.$$

These three fractions are non-negative and sum to 1 by construction.

Mittrani and Harchol-Balter use exactly this style of renewal-cycle argument and the same formula for $E[B]$ when treating busy periods and M/G/1 queues with setup times.[2][3]

3. Open Network of Exponential Servers (Jackson Network)

3.1 Model description

- There are k single-server queues (nodes), each with its own infinite buffer, FCFS discipline.[1]
- Node i has exponential service times with rate μ_i (mean $1/\mu_i$). [1]
- External arrivals to node i form a Poisson process with rate r_i (independent across nodes).
- Routing: upon service completion at node i , a job is sent to node j with probability P_{ij} or leaves the network with probability

$$P_{i,\text{out}} = 1 - \sum_{j=1}^k P_{ij}.$$

[1]

Assume the network is **open** and stable: it is possible eventually to leave the network from every node, and effective loads satisfy conditions given below.[2][3]

3.2 Traffic equations and effective arrival rates

Let λ_i be the **total arrival rate** into node i (external plus internal). Conservation of flow gives the **traffic equations**

$$\lambda_i = r_i + \sum_{j=1}^k \lambda_j P_{ji}, \quad i = 1, \dots, k.$$

In vector form, letting $\boldsymbol{\lambda} = (\lambda_1, \dots, \lambda_k)^T$, $\mathbf{r} = (r_1, \dots, r_k)^T$, and $P = [P_{ij}]$, this is

$$\boldsymbol{\lambda} = \mathbf{r} + P^T \boldsymbol{\lambda} \Rightarrow (I - P^T) \boldsymbol{\lambda} = \mathbf{r}.$$

Under the usual open-network assumptions, $I - P^T$ is invertible and this system has a unique non-negative solution.[3][2]

Define the utilization at node i as

$$\rho_i = \frac{\lambda_i}{\mu_i}.$$

The stability condition for the network is

$$\rho_i < 1, \quad i = 1, \dots, k,$$

so that each node behaves as a stable M/M/1 queue at rate λ_i, μ_i . [3]

3.3 Product-form stationary distribution

Let N_i be the number of jobs at node i in steady state, and $\mathbf{n} = (n_1, \dots, n_k)$ a generic state (joint queue lengths). For a stable **Jackson network**, the remarkable result is that the joint stationary distribution has **product form**: [2][3]

$$\pi(n_1, \dots, n_k) = \prod_{i=1}^k (1 - \rho_i) \rho_i^{n_i}, \quad n_i \in \{0, 1, 2, \dots\}.$$

Each node marginally looks exactly like an isolated M/M/1 queue with arrival rate λ_i and service rate μ_i , and the queue lengths at different nodes are independent in steady state. [3]

A standard proof uses **local balance**: for each node, the flow of jobs of that node into and out of every state is balanced, and the quasi-reversibility of M/M/1 ensures that the superposition/splitting of Poisson flows through the network preserves product form. [2]

4. M/M/1 Queue with Two Classes and Ordered State

4.1 Model description

- Single server, infinite buffer, FCFS discipline.
- Two independent Poisson arrival streams:
 - Class 1: rate λ_1 .
 - Class 2: rate λ_2 .
- Service times of all jobs are exponential with **common** rate μ (mean $1/\mu$), independent of class.[1]
- Stability: assume total load $\rho = (\lambda_1 + \lambda_2)/\mu < 1$. [3]

State description: at time t , the state is the **ordered sequence of job classes** in system, starting from the job in service and followed by jobs in the queue in FCFS order.[1]

Examples:

- Empty system: state $()$ (length 0 sequence).
- One class-2 job in system: (2) .
- Three jobs with a class-1 job in service followed by two class-2 jobs: state $(1, 2, 2)$. [1]

This defines an infinite-state CTMC on the set of all finite sequences over the alphabet $\{1, 2\}$.

4.2 Transition structure

Let a generic state be $s = (i_1, i_2, \dots, i_n)$, where each $i_k \in \{1, 2\}$ and $n \geq 0$. For convenience, denote the empty state by $\emptyset$.

Transitions are:

- **Arrival of class 1:**
 - From any state s (including empty), a class-1 arrival appends a "1" to the tail: $s \rightarrow (i_1, \dots, i_n, 1)$ at rate λ_1 .
- **Arrival of class 2:**
 - Similarly, $s \rightarrow (i_1, \dots, i_n, 2)$ at rate λ_2 .
- **Service completion:**
 - If $n > 0$, a service completion removes the head of the sequence: $(i_1, \dots, i_n) \rightarrow (i_2, \dots, i_n)$ at rate μ . If $n = 1$, this yields the empty state $\emptyset$.

Note that the service mechanism and rates do **not** depend on class; only the labels carried by arrivals differ.[3]

4.3 Idea for the stationary distribution

Let N be the number of jobs in system. Ignoring class labels, $N(t)$ evolves exactly like the customer count in a standard M/M/1 queue with total arrival rate

$$\lambda = \lambda_1 + \lambda_2, \quad \text{service rate } \mu.$$

Hence, in steady state

$$P\{N = n\} = (1 - \rho) \rho^n, \quad n = 0, 1, 2, \dots, \quad \rho = \frac{\lambda}{\mu} < 1.$$

This is the usual geometric stationary distribution for an M/M/1 queue.[2]

Now consider the **labels** of jobs. The arrival process is a merged Poisson process with rate λ , where each arriving job is class 1 with probability

$$p_1 = \frac{\lambda_1}{\lambda}, \quad p_2 = \frac{\lambda_2}{\lambda} = 1 - p_1.$$

These labels for successive arrivals are i.i.d. $\{1, 2\}$ -valued random variables with probabilities p_1, p_2 . [3]

Because service is FCFS and service times are exponential and identically distributed across classes, the service discipline is **order-preserving**: the departure order equals the arrival order, and the class labels do not influence who is served when. [3]

This suggests a candidate stationary distribution for the detailed state $s = (i_1, \dots, i_n)$:

$$\pi(s) = C \rho^n \prod_{k=1}^n p_{i_k},$$

for some normalizing constant C , where $p_{i_k} = p_1$ if $i_k = 1$ and p_2 if $i_k = 2$.

Intuitively: the factor ρ^n reflects the total number in system, and the product of p_{i_k} reflects choosing the sequence of labels as if each job

independently had class type according to the arrival mix.[3]

4.4 Normalization

Let $\mathcal{S}$ be the state space of all finite sequences over $\{1, 2\}$. For a fixed length n , the sum of probabilities of all sequences of length n under this candidate is

$$\sum_{(i_1, \dots, i_n)} \rho^n \prod_{k=1}^n p_{i_k} = \rho^n (p_1 + p_2)^n = \rho^n,$$

since $p_1 + p_2 = 1$ and the labels are independent choices.[3]

Thus the total mass is

$$\sum_{s \in \mathcal{S}} \pi(s) = C \sum_{n=0}^{\infty} \rho^n = C \frac{1}{1 - \rho}.$$

To make this sum equal 1, choose $C = 1 - \rho$. Therefore the normalized candidate is

$$\pi(i_1, \dots, i_n) = (1 - \rho) \rho^n \prod_{k=1}^n p_{i_k}, \quad n \geq 0.$$

For $n = 0$ (the empty sequence), interpret the empty product as 1, so $\pi(\emptyset) = 1 - \rho$, matching the empty-system probability of an M/M/1 queue.[2]

4.5 Verification via local balance

To confirm this is indeed the stationary distribution, check **local balance** between pairs of states that differ by a single arrival or departure.[2]

Take an arbitrary state $s = (i_1, \dots, i_n)$, $n \geq 0$, and consider an arrival of class $j \in \{1, 2\}$ that changes state from s to

$$s' = (i_1, \dots, i_n, j).$$

The transition rate from s to s' is λ_j , while the reverse transition from s' to s occurs via a service completion at rate μ .

We want to verify that

$$\pi(s) \lambda_j = \pi(s') \mu.$$

Compute each side under our candidate:

$$\pi(s) \lambda_j = (1 - \rho) \rho^n \left(\prod_{k=1}^n p_{i_k} \right) \lambda_j,$$

$$\pi(s') \mu = (1 - \rho) \rho^{n+1} \left(\prod_{k=1}^n p_{i_k} \right) p_j \mu.$$

Cancel the common factors:

$$\pi(s) \lambda_j = \pi(s') \mu \iff \lambda_j = \rho \mu p_j.$$

But

$$\rho = \frac{\lambda}{\mu}, \quad p_j = \frac{\lambda_j}{\lambda},$$

so

$$\rho \mu p_j = \frac{\lambda}{\mu} \mu \frac{\lambda_j}{\lambda} = \lambda_j.$$

Thus local balance holds for every such pair of states, which implies global balance and therefore stationarity.[2]

This argument is a standard use of local balance in product-form queueing networks and multi-class M/M/1 systems.[2][3]

4.6 Interpretation

1. Distribution of the number in system:

- Summing $\pi(s)$ over all states of length n gives $P\{N = n\} = (1 - \rho)\rho^n$, the usual M/M/1 geometric law.

2. Distribution of class sequence given $N = n$:

- Conditioning on $N = n$, the n job classes form an i.i.d. sequence with
 $P(\text{job is class 1}) = p_1, P(\text{class 2}) = p_2$.
- So in steady state, *given there are n jobs in the system, their classes and order look like taking n i.i.d. Bernoulli trials with success probability p_1* . The FCFS discipline and exponential service ensure that this labeling is consistent over time.[3]

Thus the stationary distribution of the Markov process whose state is the ordered sequence of job classes is exactly the product-form distribution derived above.

References

1. [PMCS_Assignment-1.pdf](#) - Assignment-1
2. Consider an M/M/1/1 queue with two classes of jobs. Class $i \in \{1, 2\}$
 jobs arrive acco...
3. [Harchol-Balter-Mor-Performance-modeling-and-design-of-computer-systems_queueing-theory-in-action.pdf](#) - In this chapter we saw how to model any CTMC as a DTMC where the time step in the DTMC is of length ...
4. [isi_mitrani.pdf](#) - A basic performance measure of any queueing system is the steady-state utilization, U , of its server...